$$\frac{\partial(\rho u)}{\partial x}+\frac{\partial(\rho v)}{\partial y}+\frac{\partial(\rho w)}{\partial z}=0$$

$$\rho\left(\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+w\frac{\partial u}{\partial z}\right)=-\frac{\partial p}{\partial x}+\rho g_x+\mu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}\right)$$

$$\rho\left(\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}+w\frac{\partial v}{\partial z}\right)=-\frac{\partial p}{\partial y}+\rho g_y+\mu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2}\right)$$

$$\rho\left(\frac{\partial w}{\partial t}+u\frac{\partial w}{\partial x}+v\frac{\partial w}{\partial y}+w\frac{\partial w}{\partial z}\right)=-\frac{\partial p}{\partial z}+\rho g_z+\mu\left(\frac{\partial^2 w}{\partial x^2}+\frac{\partial^2 w}{\partial y^2}+\frac{\partial^2 w}{\partial z^2}\right)$$

$$\rho\left(\frac{\partial v_r}{\partial t}+v_r\frac{\partial v_r}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta}-\frac{v_\theta^2}{r}+v_z\frac{\partial v_r}{\partial z}\right)=-\frac{\partial p}{\partial r}+\rho g_r+\mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_r}{\partial r}\right)-\frac{v_r}{r^2}+\frac{1}{r^2}\frac{\partial^2 v_r}{\partial \theta^2}-\frac{2}{r^2}\frac{\partial v_\theta}{\partial \theta}+\frac{\partial^2 v_r}{\partial z^2}\right]$$

$$\rho\left(\frac{\partial v_\theta}{\partial t}+v_r\frac{\partial v_\theta}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_\theta}{\partial \theta}+\frac{v_r v_\theta}{r}+v_z\frac{\partial v_\theta}{\partial z}\right)$$
$$=-\frac{1}{r}\frac{\partial p}{\partial r}+\rho g_\theta+\mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_\theta}{\partial r}\right)-\frac{v_\theta}{r^2}+\frac{1}{r^2}\frac{\partial^2 v_\theta}{\partial \theta^2}-\frac{2}{r^2}\frac{\partial v_r}{\partial \theta}+\frac{\partial^2 v_\theta}{\partial z^2}\right]$$

$$\rho\left(\frac{\partial v_z}{\partial t}+v_r\frac{\partial v_z}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_z}{\partial \theta}+v_z\frac{\partial v_z}{\partial z}\right)=-\frac{\partial p}{\partial z}+\rho g_z+\mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 v_z}{\partial \theta^2}+\frac{\partial^2 v_z}{\partial z^2}\right]$$

$$\frac{\partial \rho}{\partial t}+\frac{1}{r}\frac{\partial(r\rho v_r)}{\partial r}+\frac{1}{r}\frac{\partial(\rho v_\theta)}{\partial \theta}+\frac{\partial(\rho v_z)}{\partial z}=0$$

$$u=\frac{1}{2\mu}\left(\frac{\partial p}{\partial x}\right)y^2+c_1 y+c_2$$

Time to discharge from the tank using orifice at the bottom:

Circular or rectangular tank with axis vertical:

$$T=\frac{2A[H_1-H_2]}{C_d\cdot a.\sqrt{2g}}$$

Hemispherical Tank:

$$T=\frac{\pi}{C_d\cdot a.\sqrt{2g}}\left[\frac{4}{3}R\left(H_1^{\frac{3}{2}}-H_2^{\frac{3}{2}}\right)-\frac{2}{5}\left(H_1^{\frac{5}{2}}-H_2^{\frac{5}{2}}\right)\right]$$

Circular Horizontal Tank:

$$T=\frac{4L}{3C_d\cdot a.\sqrt{2g}}\left[(2R-H_2)^{\frac{3}{2}}-(2R-H_1)^{\frac{3}{2}}\right]$$

$$C_v=\frac{x}{\sqrt{4yH}};\ v=\sqrt{\frac{gx^2}{2y}}$$

$$Q=\frac{2}{3}b\sqrt{2g}\left[H_2^{\frac{3}{2}}-H_1^{\frac{3}{2}}\right]$$

$$Q=b(H_2-H_1)\sqrt{2gH}$$

$$Q=\frac{8}{15}\tan\left({}^{\theta}/_{2}\right)\sqrt{2g}\cdot H^{\frac{5}{2}}$$

Time required to empty rectangular notch = $T= 3A \times [1/(H_2)^{1/2} - 1/(H_1)^{1/2}] /[C_d \times L \times (2g)^{1/2}]$

Time required to empty triangular notch = $T = 5A \times [1/(H_2)^{3/2} - 1/(H_1)^{3/2}] / [4 \times C_d \times \tan \boldsymbol{\theta/2} \times (2g)^{1/2}]$

Still water head = $H + V_a^2 / 2g$; Francis' Formula = $Q = 1.84(L - 0.1\, nH_1)(H_1^{3/2} - h_a^{3/2})$; $H_1 = H + h_a$				
Bazin's formula = $Q = m_1(2g)^{1/2} \times L \times H_1^{3/2}$; $m_1 = [0.405 + 0.003/H_1]$; $H1 = [H + \alpha V_a^2/2g]$, $\alpha = 1.6$;				
Cipoletti notch : $Q = \frac{2}{3} C_d(2g)^{1/2} (L)[H]^{3/2}$; Broad crested weir : $Q = C_d\, L\, H_2\, (2g[H_1 - H_2])^{1/2}$; $Q_{max} = 1.7LC_d[H]^{3/2}$				
Submerged weir : $Q_1 = \frac{2}{3} C_d\, L\, (2g)^{1/2} [H_1 - H_2]^{3/2}$; $Q_2 = C_d\, LH_2\, (2g[H_1 - H_2])^{1/2}$				
$\tau = -(\boldsymbol{\partial} p/\boldsymbol{\partial} x)\, r/2$; $u = -1/4\mu\, (\boldsymbol{\partial} p/\boldsymbol{\partial} x)\, [R^2 - r^2]$; $p1 - p2/\rho g = 32\mu \bar{u} L / \rho g D^2$				
Actual momentum = $\pi\rho(-\boldsymbol{\partial} p/\boldsymbol{\partial} x)^2 R^6/48\mu^2$; Average momentum = $\pi\rho(-\boldsymbol{\partial} p/\boldsymbol{\partial} x)^2 R^6/64\mu^2$				
Actual KE= $\pi\rho(-\boldsymbol{\partial} p/\boldsymbol{\partial} x)^3 R^8/8 \times 64\mu^3$; Average KE= $\pi\rho(-\boldsymbol{\partial} p/\boldsymbol{\partial} x)^3 R^8/8 \times 128\mu^3$				
$\tau = -(\boldsymbol{\partial} p/\boldsymbol{\partial} x)\, [t - 2y]/2$; $u = -1/2\mu\, (\boldsymbol{\partial} p/\boldsymbol{\partial} x)\, [ty - y^2]$; $h_f = 12\mu \bar{u} L / \rho g D^2$				
$\mathrm{Re} = \dfrac{\rho V \ell}{\mu}$	$\mathrm{Fr} = \dfrac{V}{\sqrt{g\ell}}$	$\mathrm{Eu} = \dfrac{p}{\rho V^2}$	$\mathrm{Ca} = \dfrac{\rho V^2}{E_v}$	$\mathrm{Ma} = \dfrac{V}{c}$
$h_{L,major} = f \dfrac{L}{D} \dfrac{v^2}{2g}$	$\dfrac{1}{\sqrt{f}} = -1.8 \log_{10} \left[\left(\dfrac{\frac{\epsilon}{D}}{3.7} \right)^{1.11} + \dfrac{6.9}{Re} \right]$			
$\dfrac{D}{Dt} \int_{\mathrm{sys}} \rho d\mathit{V\!\!\!-} = \dfrac{\partial}{\partial t} \int_{\mathrm{cv}} \rho d\mathit{V\!\!\!-} + \int_{\mathrm{cs}} \rho \mathbf{V}.\hat{\mathbf{n}} dA$				
$\dfrac{D}{Dt} \int_{\mathrm{sys}} \mathbf{V} \rho d\mathit{V\!\!\!-} = \dfrac{\partial}{\partial t} \int_{\mathrm{cv}} \mathbf{V} \rho d\mathit{V\!\!\!-} + \int_{\mathrm{cs}} \mathbf{V} \rho \mathbf{V}.\hat{\mathbf{n}} dA$				
$\dfrac{\partial}{\partial t} \int_{\mathrm{cv}} (\mathbf{r} \times \mathbf{V}) \rho d\mathit{V\!\!\!-} + \int_{\mathrm{cs}} (\mathbf{r} \times \mathbf{V}) \rho \mathbf{V}.\hat{\mathbf{n}} dA = \sum (\mathbf{r} \times \mathbf{F})_{\substack{\text{contents of the} \\ \text{control volume}}}$				
$\dfrac{L}{d^5} = \dfrac{L_1}{d_1^5} + \dfrac{L_2}{d_2^5} + \dfrac{L_3}{d_3^5}$				